E17e - Fourier Analysis of Coupled Electric Oscillations

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June 10, 2025

Abstract

This report explored the resonance behavior of coupled LC oscillators and examined capacitive (low and high-point) and inductive coupling. Across the capacitvely coupled circuits it was found that fro the low-point configuration, increasing the coupling capacitance C_k reduced the coupling strength $K_{C,T}$ and an increase in coupling strength $k_{C,H}$ for the high-point configuration. Via non-linear fitted the constant capacitance across the low-point and high-point configuration was determined to be $C = (99.15 \pm 4.10)$nF and $C = (100.20 \pm 4.73)$nF respectively, in excellent agreement with the $C = 100$nF component used. The beat oscillations observed in the high-point configuration, with $C_k = 30$nF was closely aligned (4%-9% discrepancy) between measured , ($T_S = (1.2550 \pm 0.0173)$ ms,$T = (0.255600 \pm 0.024999)$ ms), and calculated periods, $T_{S,calc} \approx 1.3107$ ms and $T_{calc} \approx 0.283398$ ms, reinforcing theoretical understanding. For the inductive coupling, the mutual inductance M decayed with separation distance d following the relation $M \propto d^{-1.580 \pm 0.0968}$, the observed fractional exponent is expected to be a result of combined near/far-field magnetic influences. The results confirm and reinforced the theoretical models of LC oscillator circuits and the coupled systems.

1 Introduction

This experiment investigates the fundamental behavior of two resonantly coupled LC circuits, a classical model system frequently encountered in electronics and communication engineering. Such coupled oscillators can exchange energy, leading to distinct phenomena like frequency splitting and beat oscillations, essential for understanding resonance, signal filtering, and wireless power transfer. The circuits are examined under two different coupling conditions: capacitive coupling—where energy exchange occurs via an adjustable capacitor—and inductive coupling, achieved through spatially separated coils. The setup incorporates "high-point" and "low-point" circuits; in the high-point configuration, coupling occurs at points of maximum voltage, whereas the low-point configuration involves points of minimum voltage, affecting the coupling strength and resonance properties significantly.

In the capacitive coupling configuration, variations in coupling capacitance directly modify the frequency response of the combined circuits. In contrast, inductive coupling experiments analyze how mutual inductance—and thus the coupling strength—depends on coil separation. By carefully studying these setups, one gains insights into controlling resonance behaviors, critical for efficient energy transfer and selective signal processing. Analyzing these phenomena experimentally, supported by theoretical modeling, allows precise determination of coupling parameters and deepens the understanding of resonant circuit interactions in practical electronic applications.

2 Theoretical Analysis

In coupled LC circuits, the dynamic behavior strongly depends on the mode and strength of coupling. Capacitive coupling occurs when two resonant circuits are connected via a capacitor. In such setups, the coupling conditions are classified into "low-point" and "high-point" circuits, depending on whether the circuits are coupled at nodes of minimum or maximum voltage amplitude, respectively.

For the case of the low-point configuration, Fig.1., due to the presence of two LC circuits connected via a coupling capacitor, two distinct eigenfrequencies exists within the circuit, an in-phase (f_+) and out-of-phase (f_-). Through the use of Kirchoff's Laws these frequencies are determined to be [2]:

$$f_+ = \frac{1}{2\pi\sqrt{LC}} \quad \text{and} \quad f_- = \frac{1}{2\pi\sqrt{L\left(\frac{1}{C} + \frac{2}{C_k}\right)^{-1}}} \tag{1}$$

Where:

C_k : Coupling Capacitance (F)
C : non-Coupling Capacitance (F)
L : Inductance

With a distinct low-point coupling factor $k_{C,T}$ [2]:

$$k_{C,T} = \left|\frac{f_-^2 - f_+^2}{f_-^2 + f_+^2}\right| = \frac{C}{C + C_k} \tag{2}$$

that outlines the interaction strength between the two LC circuits as well as the energy transfer between the circuits.

In the high-point configuration, applying Kirchhoff's laws yields two distinct eigenfrequencies, corresponding to in-phase (ω_+) and out-of-phase (ω_-) oscillation modes. These frequencies depend on the coupling capacitance C_k, the circuit capacitance C, and inductance L, following [2]:

$$\omega_+ = \frac{1}{\sqrt{L(C + 2C_k)}} \quad \text{and} \quad \omega_- = \frac{1}{\sqrt{LC}}\,. \tag{3}$$

The coupling factor $k_{C,H}$, indicating coupling strength, is given by [2]:

$$k_{C,H} = \frac{\omega_-^2 - \omega_+^2}{\omega_-^2 + \omega_+^2} = \frac{C_k}{C + C_k}\,. \tag{4}$$

When only one capacitor is initially charged, beat oscillations arise due to interference between the two frequency

modes, characterized by a beat frequency $\omega_S = \omega_- - \omega_+$. The corresponding beat period T_S and the fast oscillation period T are thus directly related to the eigenfrequencies [2]:

$$T_S = \frac{2\pi}{\omega_- - \omega_+}, \quad T = \frac{2\pi}{(\omega_- + \omega_+)/2}. \tag{5}$$

For inductive coupling, two identical LC circuits interact via mutual inductance M, generated by two coils separated by a distance d. The inductive coupling yields two distinct eigenfrequencies ω_+ and ω_-, defined by [2]:

$$\omega_\pm = \frac{1}{\sqrt{C(L \pm M)}} = \frac{\omega_0}{\sqrt{1 \pm \frac{M}{L}}} \tag{6}$$

The inductive coupling factor k_L can then be expressed as the ratio of mutual inductance M to the coil inductance L [2]:

$$k_L = \frac{M}{L}. \tag{7}$$

The coupling strength typically diminishes with increasing coil separation. For large distances, the mutual inductance follows a power-law relationship $M(d) \propto d^{-n}$. Experimentally determining the exponent n provides insight into the spatial decay of magnetic fields from circular coils, it is predicted that the exponent n be 3 in accordance with traditional understanding of magnetic field decay over distance.

3 Experimental Analysis

3.1 Low-Point Circuit
(Task I)

3.1.1 Experimental Setup

Through use of a Picoscope series 5000, a DC power supply,variable capacitor, a breadboard and relevant electrical components, the coupled resonanting circuit in a low-point configuration was setup as shown in Fig.1. wherein the variable capacitor acted as the coupling capacitor C_k, with a constant voltage supplied to the circuit the coupling capacitor was varied from 80 nF to 530 nF in intervals of 50 nF for a total of 10 data points. For each data point the oscillation of the voltage across time was recording using the Picoscope 5000 series across the furthest inductor from the voltage supply. Using the Picoscope software a fourier transform was applied and the f_+ and f_- frequencies were recorded.

Through use of Eq.(2). the coupling factor $k_{C,T}$ was calculated for each data point using f_+ and f_-. The determined coupling factors were then fitted to the capacitance variant of Eq.(2). to determine the capacitance C within the circuit.

3.1.2 Measurement and Analysis

The results of the Low-Point Circuit analysis showed a clear decreasing trend in the coupling factor $k_{C,T}$ as the coupling capacitance C_k increased, Table.1. Visual Inspection of Fig.2. indicates that although the amplitude of the different f_- frequencies differ, the general trend is a convergence towards f_+, which across all C_k was consistent and non-changing. It is unknown as to whether this remains the case as C_k increases further beyond the data collected, nor if the f_- does converge to f_+ or if they both converge to a different frequency such as the complete circuit's resonant frequency f_0.

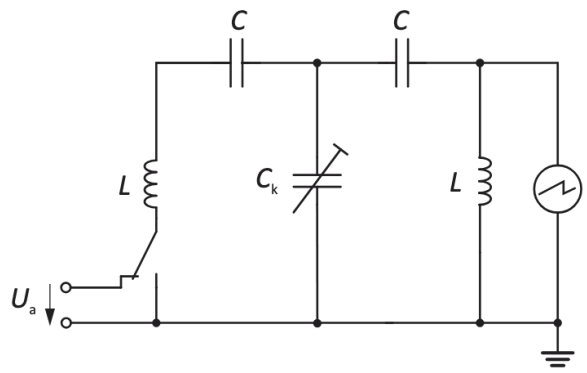


Figure 1: Low-Point Circuit of Capacitively Coupled Resonant Circuits

Source: [2]

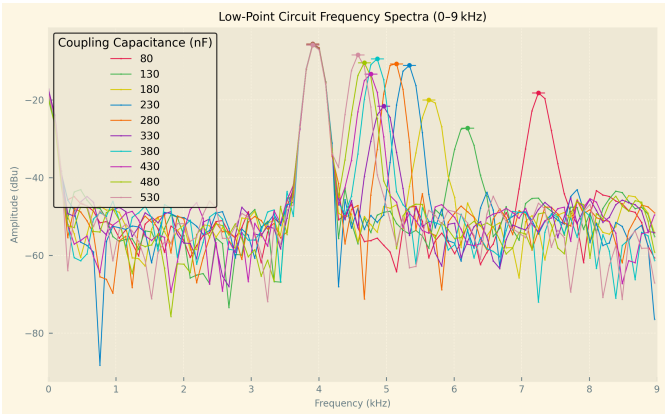


Figure 2: Frequency Spectra of all coupling capacitance for the Low-Point Circuit

C_k (nF)	f_+ (kHz)	f_- (kHz)	$k_{C,T}$
80	3.91 ± 0.095	7.25 ± 0.095	0.549 ± 0.0194
130	3.91 ± 0.095	6.20 ± 0.095	0.431 ± 0.0235
180	3.91 ± 0.095	5.63 ± 0.095	0.349 ± 0.0261
230	3.91 ± 0.095	5.34 ± 0.095	0.302 ± 0.0275
280	3.91 ± 0.095	5.15 ± 0.095	0.269 ± 0.0284
330	3.91 ± 0.095	4.96 ± 0.095	0.233 ± 0.0294
380	3.91 ± 0.095	4.86 ± 0.095	0.215 ± 0.0298
430	3.91 ± 0.095	4.77 ± 0.095	0.196 ± 0.0303
480	3.91 ± 0.095	4.67 ± 0.095	0.176 ± 0.0308
530	3.91 ± 0.095	4.58 ± 0.095	0.156 ± 0.0313

Table 1: Measured values of f_+, f_-, and $K_{C,T}$ for different capacitances C.

Comparing the results of Table.1. and Fig. 3. indicates that the decreasing trend is such that $k_{C,T} \propto C_k^-1$ inline with the fitting equation Eq.2. used, resulting in a strong fitting with $r^2 = 0.9980$ and excellent agreement with theoretical analysis. The fitting predicts a capcitance of $C = (99.15 \pm 4.0977)$nF largely inline with experimentally used 104K capacitors, $(100 \pm 10\%)$nF, indiviudally measured at 102.3nF and 104.3nF.

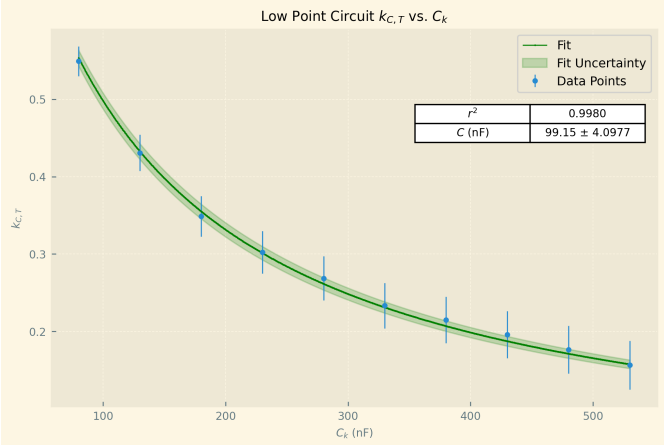


Figure 3: Fitted Curve of Coupling Capacitance C_k vs. Coupling Factor $k_{C,T}$

3.2 High-Point Circuit
(Task II & III)

3.2.1 Experimental Setup

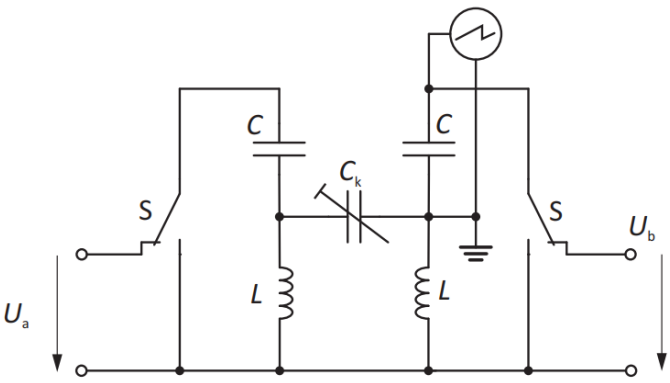


Figure 4: High-Point Circuit of Capacitively Coupled Resonant Circuits

Source: [2]

In this setup, two LC circuits were capacitively coupled in the high-point configuration, as illustrated in Fig.4. Each LC resonator consisted of an inductor and capacitor arranged such that the coupling capacitor C_k connects the circuits at points of maximum voltage amplitude ("high points"). The coupling capacitor C_k was varied systematically from 30 nF to 130 nF, and for each value, a waveform generator (Picoscope series 5000) provided an initial excitation. The resulting oscillations were recorded using the digital oscilloscope, and their frequency spectra were obtained by performing a Fast Fourier Transform (FFT) analysis, allowing identification of the characteristic coupled-mode frequencies f_+ and f_-.

To specifically examine beat oscillations in the time domain, minor modifications were introduced to the setup in Fig. 4. One resonant circuit was charged initially, while the second circuit remained uncharged by short-circuiting its switch. After initiating oscillations by closing the charged circuit's switch, energy transferred periodically between the two circuits, creating observable beats. The oscillation voltage was captured with the oscilloscope as a function of time, providing a time-domain record from which both the fast oscillation period T and the beat envelope period T_S were determined.

3.2.2 Measurement and Analysis

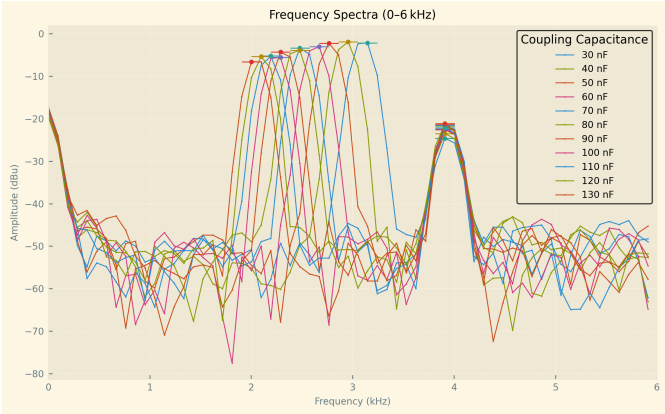


Figure 5: Frequency Spectra of High-Point Circuit with 10 Different Coupling Capacitances C_k

The frequency spectra were recorded for various coupling capacitances C_k ranging from 30 nF to 130 nF. For each spectrum, the in-phase and out-of-phase resonance frequencies f_+ and f_- were identified by performing peak detection on the FFT data. The coupling factor $k_{C,H}$ was then calculated from these measured frequencies using the relation:

$$k_{C,H} = \frac{f_-^2 - f_+^2}{f_-^2 + f_+^2} .$$

To determine the system's inherent capacitance C, the measured values of $k_{C,H}$ were fitted to the theoretical model provided by Eq. (4).

C_k (nF)	f_+ (kHz)	f_- (kHz)	$k_{C,H}$
30	3.15 ± 0.095	3.91 ± 0.095	0.214 ± 0.0371
40	2.96 ± 0.095	3.91 ± 0.095	0.273 ± 0.0374
50	2.77 ± 0.095	3.91 ± 0.095	0.333 ± 0.0376
60	2.67 ± 0.095	3.91 ± 0.095	0.364 ± 0.0375
70	2.48 ± 0.095	3.91 ± 0.095	0.426 ± 0.0373
80	2.48 ± 0.095	3.91 ± 0.095	0.426 ± 0.0373
90	2.29 ± 0.095	3.91 ± 0.095	0.490 ± 0.0367
100	2.29 ± 0.095	3.91 ± 0.095	0.490 ± 0.0367
110	2.19 ± 0.095	3.91 ± 0.095	0.521 ± 0.0363
120	2.10 ± 0.095	3.91 ± 0.095	0.553 ± 0.0358
130	2.00 ± 0.095	3.91 ± 0.095	0.584 ± 0.0352

Table 2: Measured values of f_+, f_-, and $K_{C,H}$ for different capacitances C_k.

The fitting process yielded a capacitance value of $C = (100.20 \pm 4.73)$ nF. This experimental result is in good agreement with the separately measured capacitances of the two LC circuits, which were 102.5 nF and 104.3 nF, both of which lie within the uncertainty of the obtained value. The large value of the correlation coefficient $r^2 = 0.9864$, demonstrates excellent agreement between the experimental data and the theoretical model The slight deviation between the obtained C and the individually measured values can be attributed to component tolerances and minor parasitic capacitances within the measurement setup. Overall, the experimental

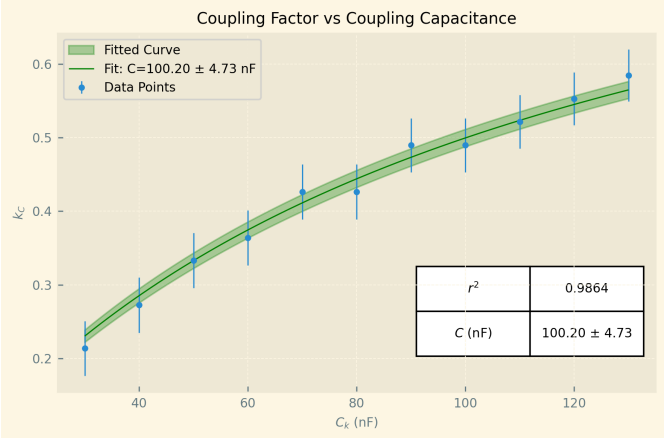


Figure 6: Fitted Curve of Coupling Capacitance C_k vs. Coupling Factor k_C

data closely match theoretical expectations, confirming the validity of the coupling model.

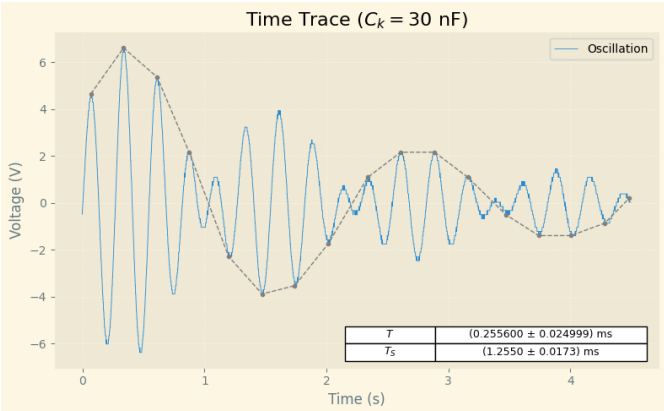


Figure 7: Time Trace Using the Coupling Capacitor $C_k = 30$ nF

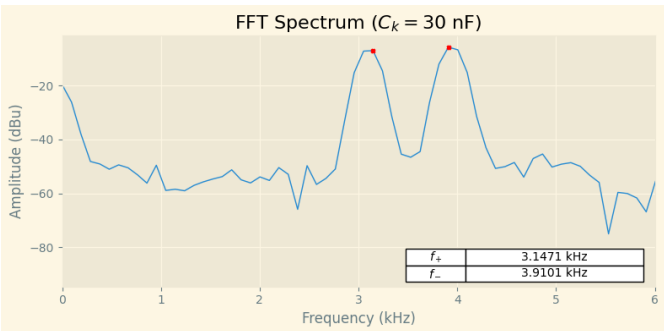


Figure 8: Time Trace Using the Coupling Capacitor $C_k = 30$ nF

To observe the beat oscillations in the high-point circuit, a coupling capacitor of $C_K = 30\,\text{nF}$ was used. From the recorded time trace in Fig. 7, the beat period was measured directly as $T_S = (1.2550 \pm 0.0173)\,\text{ms}$, and the oscillation period was found to be $T = (0.255600 \pm 0.024999)\,\text{ms}$. Simultaneously, the frequency spectrum of the signal was obtained through Fourier analysis. This spectrum clearly revealed two distinct peaks corresponding to the two eigenmodes of the coupled oscillation. Specifically, the measured in-phase frequency f_+ was $(3.1471 \pm 0.0954)\,\text{kHz}$, while the out-of-phase frequency f_- was $(3.9101 \pm 0.0954)\,\text{kHz}$.

To verify the consistency of these results, the oscillation and beat periods were also calculated independently using these measured frequencies. The theoretical oscillation period, computed as the reciprocal of the mean of the two frequencies given by the expression in (5), was $T_{\text{calc}} = 0.283398\,\text{ms}$. Similarly, the theoretical beat period, obtained from the inverse of the frequency difference, yielded $T_{S,\text{calc}} = 1.3107\,\text{ms}$.

Comparing these theoretical calculations with the direct measurements, good agreement was observed, with discrepancies on the order of approximately 4–9%. Such discrepancies can reasonably be attributed to experimental uncertainties, including systematic errors in the measurement of time intervals, inherent noise in the signal, and damping effects that were neglected in the ideal theoretical model. The consistency between the two methods underscores the validity of interpreting the beat oscillations as interference patterns of two coupled resonant frequencies. The frequency spectrum thus provided essential cross-validation and enhanced confidence in the measurement reliability.

3.3 Inductively Coupled Resonant Circuits
(Task IV)

3.3.1 Experimental Setup

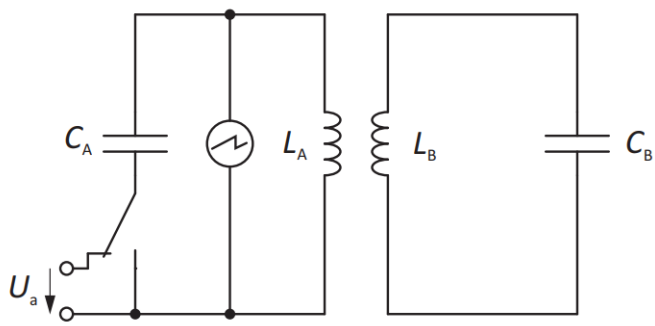


Figure 9: Inductively Coupled Resonant Circuits

Source: [2]

for the inductive coupling experiment, the setup used two identical coils placed coaxially facing each other separated by a distance d. The circuits were constructed as outlined in Fig.9. with Circuit A depicting primary circuit that is supplied a interim DC voltage, with use of the Picoscope 5000 Series the voltage across time is also measured across the Circuit A inductor L_A. Each coil, paired with its capacitor, formed an LC resonator. To calculate the coupling strength k_L between the two circuits voltage is supplied and the voltage profile and frequency spectra is measured with the Picoscope. This was repeat for a total of 8 measurements across a range of $d \in [4, 11]$cm in intervals of 1cm. After which Circuit B was placed sufficiently far away and the resonant frequency f_0 of Circuit A was recorded to determine the the inductance L of the two identical coils via:

$$L = \frac{1}{4\pi^2 f_0^2 C}$$

To determine the k_L Eq.(6). and Eq.(7). were used to

arrive at the relationship:

$$k_L = \frac{\omega_-^2 - \omega_+^2}{\omega_-^2 + \omega_+^2} \tag{8}$$

in order to calculate k_L for each separation distance d. Finally with determined k_L and L and through Eq.7 a logarithmic linear fitting of the form:

$$\log_{10}(M) = \log_{10}(A) - n\log_{10}(d)$$

was then performed to analysis the relation $M \propto d^{-n}$ and determine a relation for n. This determined n was then compared to previously explored work [1].

3.3.2 Measurement and Analysis

The determined resonant frequency for the Inductive Coupling circuit was $f_0 = (3.4060.018)$KHz, leading to a determined inductance of $L = (21.8380.226)$ mH. Comparing this result to the trend seen in Fig. 10., it can be seen that as the separation distance d increases f_- and f_+ appear to converge to f_0 as the inductive influence of Circuit B on Circuit A diminishes. It can also be seen

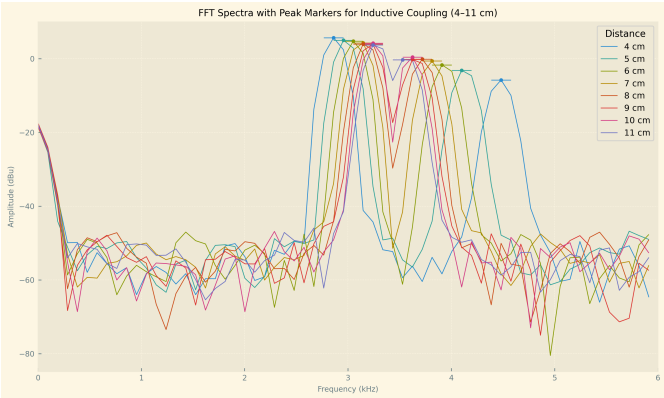


Figure 10: Frequency Spectra for all separation distances of the Inductively Coupled Resonant Circuits

visually from Fig. 10. and quantitatively from Table.3. that the out-of-phase frequency f_- diminishes at a faster rate than the increase of the in-phase frequency f_+ as d increases. Table.3. also shows the experimentally gathered coupling strengths k_L, determined via Eq.(8)., show a rapidly decreasing trend in k_L as separation distance d increases, indicating that the inductive strength between the two inductive coils rapidly decreases as separation distances increases linearly.

d (cm)	f_+ (kHz)	f_- (kHz)	k_L
4	2.86 ± 0.095	4.48 ± 0.095	0.421 ± 0.0325
5	2.96 ± 0.095	4.10 ± 0.095	0.316 ± 0.0360
6	3.05 ± 0.095	3.91 ± 0.095	0.243 ± 0.0373
7	3.15 ± 0.095	3.81 ± 0.095	0.190 ± 0.0379
8	3.15 ± 0.095	3.72 ± 0.095	0.166 ± 0.0386
9	3.24 ± 0.095	3.62 ± 0.095	0.111 ± 0.0390
10	3.24 ± 0.095	3.62 ± 0.095	0.111 ± 0.0390
11	3.24 ± 0.095	3.53 ± 0.095	0.084 ± 0.0397

Table 3: Measured values of f_+, f_-, and k_L for different distances d for the inductively coupled resonant circuits.

Fig. 11. expands on this observation with a determined inverse exponent $n = 1.580 \pm 0.0968$ meaning that $M = Lk_L \propto d^{-1.580}$, this fractional exponent is initially curious and is expected to be a results of combined near and far-field magnetic influences however evaluation of this was not performed.

The determined n was compared with previous work on magnetic coil influences, wherein the relation between $B \propto d^{-n}$ of the trailing end of a Helmholtz Coils configuration of a similar distance range $d \in [4-10]cm$ from the closest coil. n for this previous work was determined to be $n_{E7} = 1.4070.101$ which is comparable to the $n = 1.580 \pm 0.0968$ determined in this report. Although these number agree with each other within experimental uncertainty, it is unknown as to whether the presence of the fractional exponent ≈ 1.5 is a result of combined near and far-field magnetic influences or whether $n \approx 1.5$ is a theoretically sound result for these particular cases.

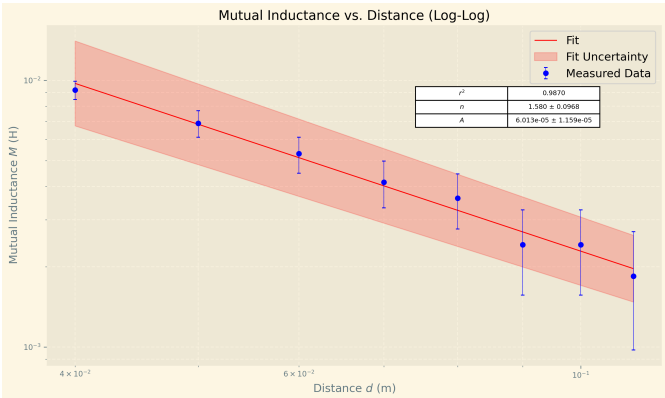


Figure 11: Fitted Curve of Separation Distance d and Mutual Inductance M on a Logarithmic Scale

4 Discussion

The experimental investigation into the capacitor coupled LC Circuit in a Low-Point configuration yielded results that are consistent with theoretical predictions. As the coupling capacitance C_k increased from 80 nF to 530 nF, the in-phase frequency (f_+) was remarkably stable at (3.91 ± 0.095)KHz across the values of C_k, while the out-of-phase (f_-) frequency decrease from (7.25 ± 0.095)KHz to (4.58 ± 0.095)KHz. This experimental behavior aligns with the theoretical analysis and Eq. 1, where f_+ is dependent on non-changing variables C and L, and f_- is dependent on an inverse relationship with C_k. Furthermore the relationship reinforces the observed trend,Fig.2., that f_- converges to f_+ as C_k increases, outlining that as $C_k \to \infty$ $f_- \to f_+ = f_0$, outlining that when coupling capacitance is sufficiently large the two different frequencies across the circuit, become a single resonant frequency.

the convergence of $f_- \to f_+$ as C_k increases is inherit in the coupling factor $k_{C,T}$ Eq.2. as the numerator tends to 0 as C_k increases. Again this theoretical trend is reflected in the experimental results Fig.3. of a clear decreasing,inverse relationship, from 0.549 to 0.156, which is in strong agreement,($r^2 = 0.9980$), with the fitted equation Eq.2 for a circuit capacitance $C = (99.15 \pm 4.10)$nF which aligns perfectly with the 104K,$(100 \pm 10\%)$nF capacitors used. This trend again shows that for sufficiently high

coupling capacitance's the two different frequencies across the circuit, become a single frequency, and the coupling strength diminishes.

the analysis of the frequency spectrum and the coupling factor of the Low-Point LC resonating circuit the results expertly demonstrated the theoretical models and expected results, the determined capacitance $C = (99.15 \pm 4.10)$nF is well-within the uncertainty of the 104K capacitors used, however slightly deviates from the individually measured results of 102.3 nF and 104.3 nF however this deviation along with other slight deviations are likely a results of component tolerances, slight parasitic effects and chosen sample rates in regards to the frequency measurements. For further investigations it is recommended to investigate a larger range of C_k to explore the convergence of the f_- and f_+ frequency.

In the capacitive coupling experiments conducted using the high-point configuration, the results consistently aligned well with theoretical expectations. As the coupling capacitance, C_k, was varied from 30 nF to 130 nF, distinct in-phase and out-of-phase frequencies, f_+ and f_-, were clearly identified. The coupling factor, $k_{C,H}$, calculated from these frequencies, demonstrated a decreasing trend as C_k increased, accurately fitting the theoretical model provided by Eq. (4). The determined intrinsic circuit capacitance of $C = (100.20 \pm 4.73)$ nF closely matched the separately measured capacitances (102.5 nF and 104.3 nF), demonstrating the reliability of the experimental setup. Small deviations observed could be reasonably attributed to tolerances in component values, minor parasitic capacitances, and measurement uncertainties.

The direct measurement of the beat period $T_S = (1.2550 \pm 0.0173)$ ms and the oscillation period $T = (0.255600 \pm 0.024999)$ ms agreed notably well with theoretically calculated periods derived from frequency measurements, $T_{S,calc} \approx 1.3107$ ms and $T_{calc} \approx 0.283398$ ms, respectively. Observed discrepancies were within acceptable limits (approximately 4–9%), reflecting reasonable systematic errors including damping effects, inherent noise in signal acquisition, and precision limits of the measurement instruments.

The consistency between time-domain and frequency-domain analyses reinforced the theoretical interpretation of beat oscillations as interference between two eigenmodes with distinct resonant frequencies. The frequency domain data provided a critical cross-validation mechanism, confirming the robust and accurate capture of resonant behavior in the coupled circuit. Future studies could explore additional factors such as damping mechanisms and a broader range of coupling capacitances to further elucidate the dynamics of coupled resonant circuits.

The investigation into the inductively coupled LC circuits showed a convergent trend of f_- and f_+ towards the uncoupled resonant frequency $f_0 = (3.406 \pm 0.018)$KHz as separation distance d increases with f_- converging at a faster rate than f_+,Fig.10. The determined k_L values, Table.3, showed a rapidly decreasing inductive coupling strength as d increased linearly with is consistent with theoretical expectations of magnetic interactions however a fitted analysis, Fig 11, showed a power-law relation ship between $M = Lk_L \propto d^{-n}$ with $n = 1.580 \pm 0.0968$ deviating from an expected theoretical relation of $n \approx 3$ for magnetic dipole fields. A similar fractional exponent was also observed in a previous analysis into magnetic fields in coils, where the magnetic field followed the relationship $B \propto d^{-n}$ with $n_{E7} = 1.407 \pm 0.101$ at distances $d \in [4, 10]$cm from a Helmholtz Coil configuration. The observation of a fractional exponent is believed to arise from measurements taken within a range that includes both the near-field dominated and then far-field dominated magnetic influences, however this was not validated.

5 Conclusion

This experiment provided an in-depth analysis of coupled resonant LC circuits with both capacitive and inductive coupling, using low-point and high-point configurations. These setups model real-world systems used in signal processing, wireless communication, and power transfer, making their study crucial for understanding how energy and information are transmitted in modern electronic devices. In the low-point setup, increasing the coupling capacitance caused the out-of-phase frequency to decrease and approach the in-phase frequency, a behavior that matched theoretical expectations. This convergence confirms the accuracy of the underlying model. The extracted circuit capacitance, $C = (99.15 \pm 4.10)$ nF, was in strong agreement with the nominal component values, indicating a high degree of consistency between theory and experiment.

Similarly, the high-point circuit results aligned well with predictions. As the coupling capacitor C_k increased, the calculated coupling factor $k_{C,H}$ decreased, as expected from the theoretical model. The fitted capacitance value, $C = (100.20 \pm 4.73)$ nF, closely matched the separately measured component values, reinforcing the reliability of the measurement approach.

The analysis of beat oscillations using a 30 nF coupling capacitor demonstrated strong agreement between time-domain measurements and theoretical calculations. The measured beat period $T_S = (1.2550 \pm 0.0173)$ ms and oscillation period $T = (0.255600 \pm 0.024999)$ ms were within 4–9% of the values calculated from the measured eigenfrequencies. These small deviations can be reasonably attributed to common experimental uncertainties, such as measurement resolution, signal noise, and unaccounted damping.

The investigation into the inductive coupling experiments a relationship between mutual inductance and separation distance of $M \propto d^-n$ with $n = 1.580 \pm 0.0968$ over a range $d \in [4 - 11]$cm, an un-intuitive result considering theoretical understanding of magnetic relations typically being $B \propto d^{-3}$, however this result was consistent with prior Helmholtz Coil measurements, ($n_{E7} = 1.407 \pm 0.101$) over a similar distance range, suggesting combined near and far-field influences may be the result of an observed fractional exponent. Additionally an asymmetric convergence of in-phase f_+ and out-of-phase f_- was observed as d increases with f_- converging at a faster rate than f_+ as both converging to an uncoupled resonant frequency of $f_0 = (3.406 \pm 0.018)$KHz. Future investigation should explore a larger and more density separation range to evaluate the relation power-law relation between $M \propto d^{-n}$ throughout defined near-field and far-field distances.

Overall, the experiment confirmed major theoretical results regarding coupled resonant systems, especially the behavior of mode splitting, beat formation, and coupling strength variation. It also highlighted how frequency-domain and time-domain analyses can complement each other in characterizing complex oscillatory behavior. Future work could explore the influence of damping or nonlinear components, which may reveal how energy dissipation or amplitude-dependent effects modify the resonance behavior and stability of the system. For instance, introducing resistive elements or nonlinear capacitors could simulate real-world imperfections and broaden the understanding of system robustness.

References

[1] B. C. Grey, J. *E7e Magnetic Fields in Coils.* Universität Leipzig, Novemember 2024.

[2] M. Ziese. *E17e Fourier Analysis of Coupled Electric Oscillations.* Universität Leipzig, June 2025.

Appendix: Preparation Material

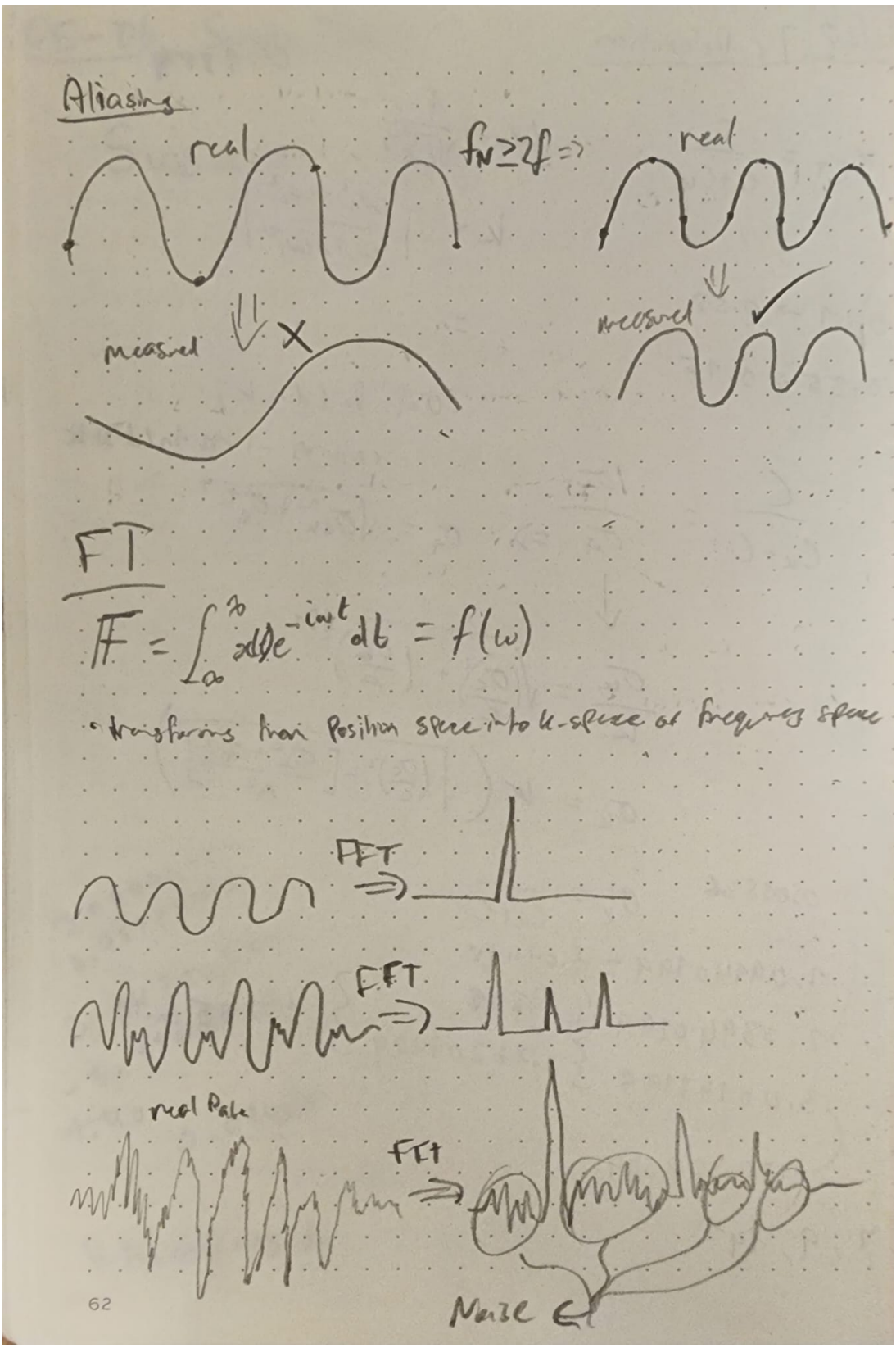


Figure 12: Preparation Material covering; Aliasing, Nyquist-Shannon Theorem, Frequency Spectra and the Fourier Transformation

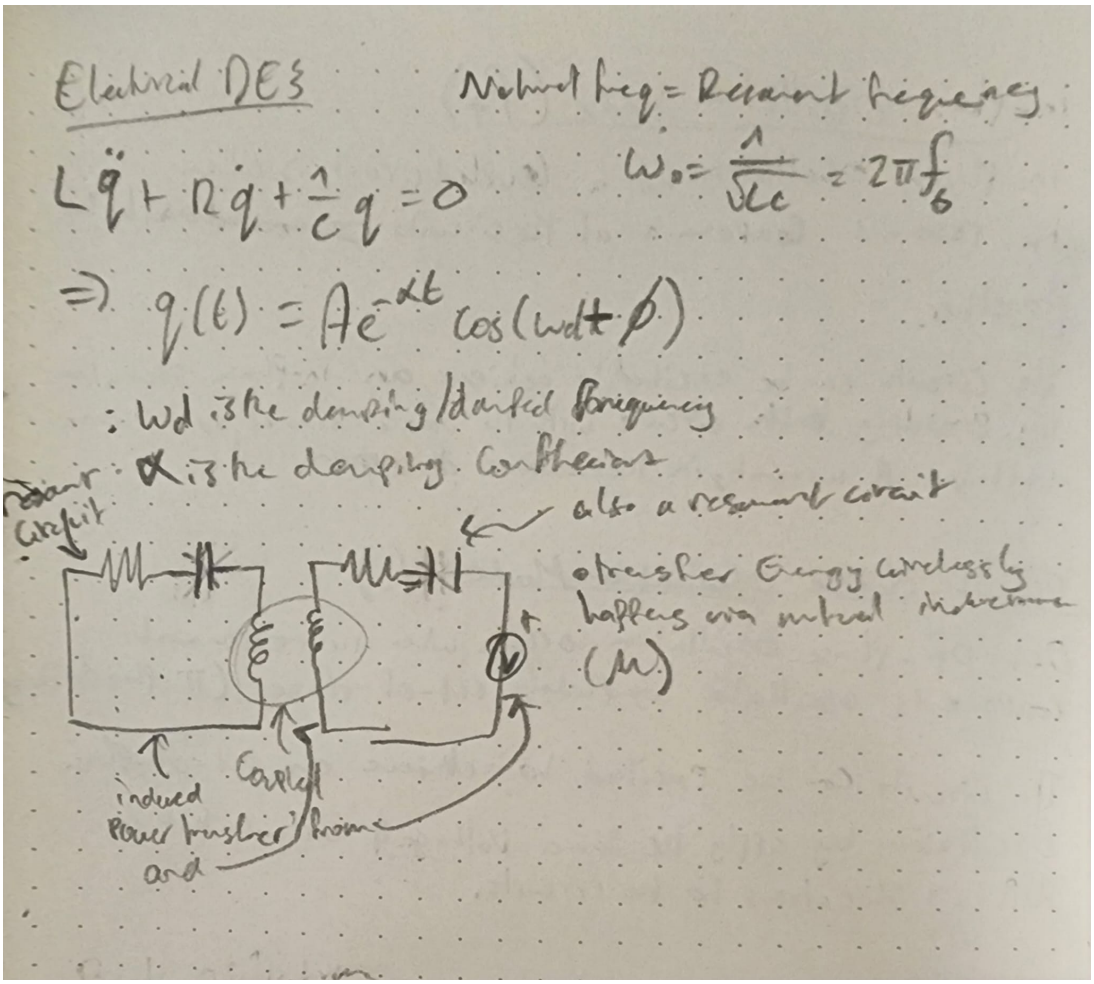
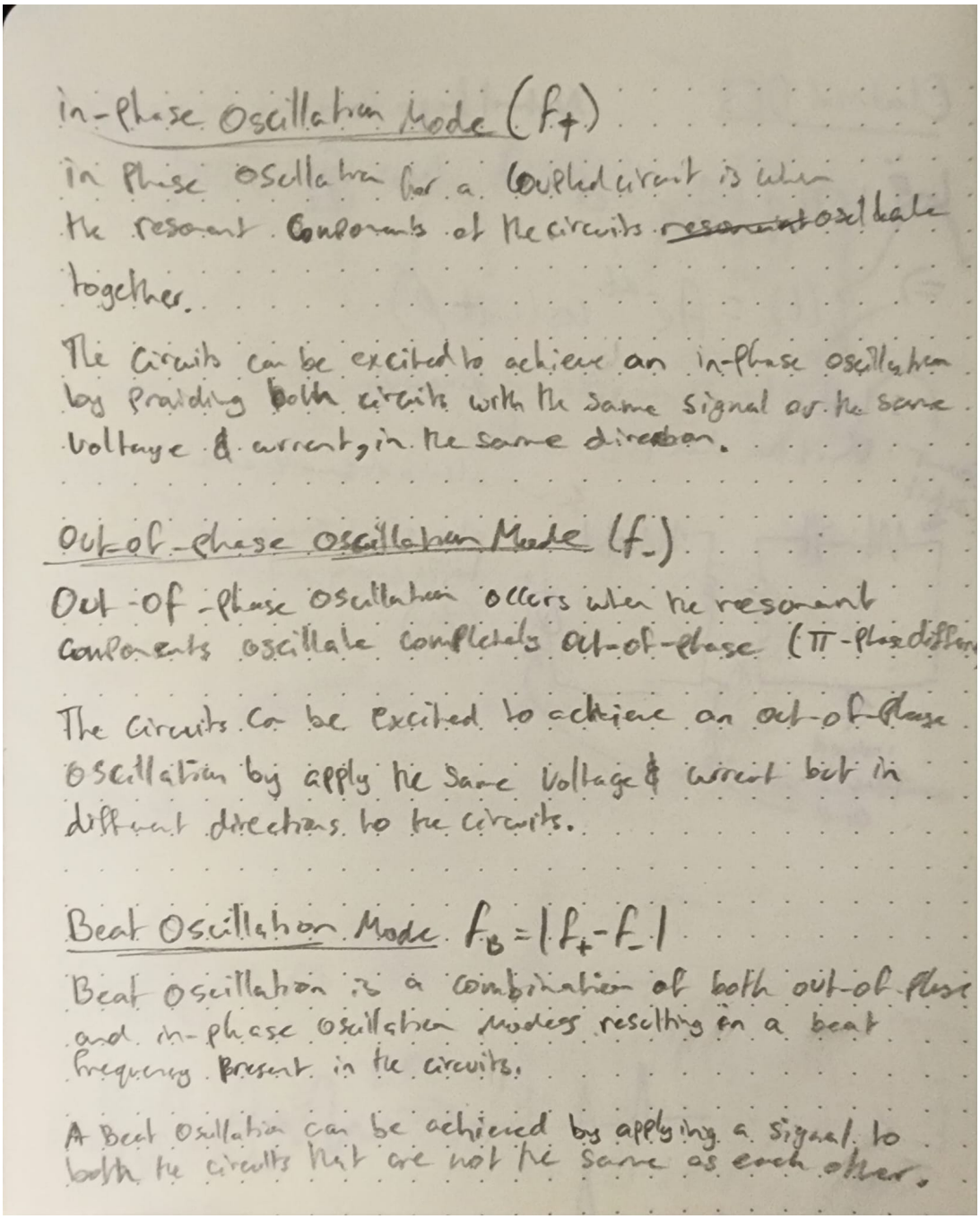


Figure 13: Preparation Material covering; non-/Coupled Resonant Circuits Differential Equations of damped electrical oscillations there solutions, as well as natural frequency (resonant frequency)



In-Phase Oscillation Mode (f_+)

In Phase oscillation for a coupled circuit is when the resonant components of the circuits ~~resonant~~ oscillate together.

The circuits can be excited to achieve an in-phase oscillation by providing both circuits with the same signal or the same voltage & current, in the same direction.

Out-of-phase Oscillation Mode (f_-)

Out-of-phase oscillation occurs when the resonant components oscillate completely out-of-phase (π-phase differ

The circuits can be excited to achieve an out-of-phase oscillation by apply the same voltage & current but in different directions to the circuits.

Beat Oscillation Mode $f_B = |f_+ - f_-|$

Beat oscillation is a combination of both out-of-phase and in-phase oscillation modes resulting in a beat frequency present in the circuits.

A Beat oscillation can be achieved by applying a signal to both the circuits that are not the same as each other.

Figure 14: Preparation Material covering; In-Phase, Out-of-Phase and Beat Oscillations and how to induce/excite them in resonant circuits